Dr. Lala Kounta

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ACADEMIC APPOINTMENT

2024 - now **Research associate**, Institute for Biodiversity, Ecology, Evolution, and Macrosystems, Department of Integrative Biology, Michigan State University (East Lansing, Michigan, USA)

2023 - 2024 **Post-doctoral fellow**, Institute for Biodiversity, Ecology, Evolution, and Macrosystems, Department of Integrative Biology, Michigan State University (East Lansing, Michigan, USA)

2022-2023 **African Future Leadership Fellow**, Alliance for African Partnership, Department of Integrative Biology, Michigan State University (East Lansing, Michigan, USA)

2020-2022 **Lecturer** Cheikh Anta Diop University (UCAD; Dakar, Senegal), Gaston Berger University (Saint Louis, Senegal).

Jul 2021- Mar 2022. **Visiting Scientist**, Oceanic Platform of the Canary Islands (PLOCAN), (Canaries, Spain)

Nov20- **Visiting Scientist**, African Institute of Mathematical Sciences Rwanda (AIMS), (Kigali, Rwanda)

EDUCATION

2019

- **PhD in Environmental Sciences**
  Sorbonne University (Paris, France)
- **PhD in Physical Oceanography and Climate Sciences**
  Cheikh Anta Diop University (Dakar, Senegal)

2013 **MS of Research in Meteorology, Oceanography, and Land Management**
UCAD, Dakar, Senegal

2012 **MS in Physics**
UCAD, Dakar, Senegal

2011 **BS in Physics and Chemistry**
UCAD, Dakar, Senegal

RESEARCH PROJECTS

2025-2026 **In the context of extreme events and biotic interactions, how will solar radiation modification (SRM) alter ecological structure and function?** funding by the Environmental Defense Fund, (Role: co-**Principal Investigator with Phoebe Zarnetski, Alisha Shah, Kelly Kasper, Luo Lifeng**)

2024 - Now: Monitoring Senegalese Coast for Sustainable Ocean' funding by the World Academy of Sciences, UNESCO, (Role: **Principal Investigator**), Senegal

2023 - Now	**Investigating potential ecological impacts of climate intervention by reflecting sunlight to cool Earth**', funded by the National Science Foundation (role: **research scientist**), MSU, USA
2023- 2024	**Research to Inform Inclusive Management of the Cayar and Ufoyaal Kassa-Bandial MPAs in Senegal**, funded by Lenfest Ocean Program (role: **key personal**), Senegal
2023-2024	**Risks and vulnerability of the Senegalese marine and coastal environment: characterization and future evolution**, funding by British Petroleum (role**: key personal**), Senegal
2022-2023	**Characterizing the role of interannual environmental variability in structuring the life histories of global ecological communities** (Role: **co-Principal Investigator** with K. Kapsar, C. Youngflesh , A. Uscanga, P. William) funded by the Institute for Biodiversity, Ecology, Evolution, and Macrosystems, MSU, USA
2017-2019	SCUS-2050: Exploring the future of the Southern Canary Upwelling System' funded by Future climate for Africa (**co-Principal Investigator**), **Senegal**

TEACHING

2025 **Spatial Ecology, (Lab Instructor),** College of Natural Science, MSU
https://space-lab-msu.github.io/MSUGradSpatialEcology/index.html

2018-2022 **Probability & statistics: instructor**
Ecole Supérieure Polytechnique/ Cheikh Anta Diop University, Senegal

2018-2022 **Physical oceanography: instructor**

- Ecole Supérieure Polytechnique / Cheikh Anta Diop University, Senegal
- Institut Universitaire de pêche et Aquaculture (IUPA), Cheikh Anta Diop University
- UFR des Sciences Agronomes de l'Aquaculture et des Technologies Alimentaires / Gaston Berger University

2019-2022 **Ocean modelling: instructor**
Institut Universitaire de pêche et Aquaculture, UCAD, Senegal

2019-2022 **Thermodynamic: instructor**
Ecole supérieure des Génis des Procédés industriels , Senegal

2020-2022 **Costal climate and climate variability : instructor**
Faculté des Lettres et Sciences Humaines, UCAD, Senegal

AWARDS AND GRANTS

2025-2026	**Environmental Defense Fund (**Role: co-**Principal Investigator with Phoebe Zarnetski, Alisha Shah, Kelly Kasper**	**$264,433**
2025-2026	**World Academy of Sciences, UNESCO**, (role: **Principal Investigator)**	**$68,390**
2023- 2024	**Institute for Biodiversity, Ecology, Evolution, and Macrosystem, MSU(**Role: **Fellow** with K. Kapsar, C. Youngflesh, A. Uscanga, P. William)	**$75,000**
2022-2023	**African Futures Research Leadership Program Fellowship**, Michigan State University (role: **Fellow**)	**$50,000**
2022-2024	**Faculty for the Future Program Fellowship**	**$60,000**

TRAINING RECEIVED: PROFESSIONAL DEVELOPMENT, SKILLS, AND DEI

2023

- Women of color in leadership, MSU
- Equitable Partnerships in Global Research and Innovation, MSU
- Diversity, Equity, and Inclusion: Cross-cultural Communication
- Grant Proposal Development Workshop

2022

- Developing a supportive mentoring network for strategic career advancement
- Academic time management, MSU

2020

- The Canary Current Eastern Boundary Upwelling System, Cabo Verde
- Climate Change science, Rwanda

2018 Hydrodynamic and Environmental Modeling
2017 Geographic Information Systems (GIS): Application in the Marine Sector
2016

- Geophysical Data Processing and Statistics
- Fundamental Principles of Oceanographic Data Management

OUTREACH & ENGAGEMENT

2025- Now ***Cheikh Anta Diop University***
Financial and mentoring support: Supported top-performing female students in marine sciences through my UNESCO-TWAS-funded program, implemented in partnership with IUPA to strengthen gender equity and academic excellence (Dakar, Senegal)

2023 Environmental Data Science Innovation & Inclusion Lab (ESIIL) Innovation Submit
Mentoring: Mentored a diverse group of participants, including early career researchers, Indigenous communities, and professionals by offering advice, feedback, and making connections (Boulder, Colorado, USA)

2020 Girls' Middle School in Rwanda
Guest Speaker: Gave a talk on talk on Women in Science (Kigali, Rwanda)

SKILLS

Statistical data analysis, numerical modeling, machine learning, data cleaning, data analysis, big data analysis, data visualization, Cloud computing, high-performance computing, MS Office, critical thinking, problem-solving, teamwork, Oral Communication, Cultural Competency, report writing,

Computing Program & Language: **Python, R, MATLAB, SQL, NetCDF, CDO**

MEDIA COVERAGE

Life in the fast, or slow, lane: How environmental variability influences birds' survival and reproduction strategies, February, 2025
African Women. Achieving one's goals**,** June, 2025
La mujer que quiere tomarle la temperatura al mar, elpais, newsletter, 2022
Ocean temperature increase along senegalese coast could reduce sardine fisheries, Future climate for Africa News, 2019
IRDTalents : rencontre avec Lala Kounta, titulaire d'une bourse pour une thèse au sud (ARTS), IRD, 2017

PUBLICATIONS

Under review

Lala kounta, Lifeng Luo, Gouri Anil, Daniel M. Hueholt, Cheryl Shannon Harrison, Daniele Visione, Mari Rachel Tye, Tyler Felgenhauer, Amadou T Gaye and Phoebe L Zarnetske. Climate Intervention

through Stratospheric Aerosol Injection may partially mitigate marine heatwave https://doi.org/10.22541/essoar.176442822.27571009/v3 (preprint)

Kapsar, K., **L. Kounta L**.,P. Bills,P., S. Record, A. Strecker, B**.** Baiser, B., PL Zarnetske, Multi-scale geodiversity and climate data for the National Ecological Observatory Network (NEON). **Data paper**

Peer Reviewed

Youngflesh, C., Kapsar, K., Uscanga, A., Williams, P.J., Doser, J.W., **Kounta, L**. and Zarnetske, P.L. (2025), Environmental Variability Shapes Life History of the World's Birds. Ecology Letters, 28:e70077. https://doi.org/10.1111/ele.70077

Kounta, L. 2019. Le rôle de la dynamique océanique et atmosphérique en Atlantique Nord sur le fonctionnement de l'upwelling ouest-africain. PhD Thesis, Sorbonne Université / Université Cheikh Anta Diop (Dakar), https://theses.hal.science/tel-03363982. (**PhD manuscript**)

L. Kounta, X. Capet, J. Jouanno, N. Kolodziejczyk, B. Sow, and A. Thierno Gaye. 2018. A Model Perspective on the Dynamics of the Shadow Zone of the Eastern Tropical North Atlantic – Part 1: The Poleward Slope Currents along West Africa. Ocean Science 14, no. 5: 971–9https://doi.org/10.5194/os-14-971-2018

Ndoye, S., X. Capet, P. Estrade, E. Machu, **L. Kounta**, B. Sow, M. Diakhaté, and A. T. Gaye. 2018. A Numerical Modeling Study of the Southern Senegal Upwelling Shelf: Circulation and Upwelling Source Waters. African Journal of Environmental Science and Technology 12, no. 12: 487–500. https://doi.org/10.5897/AJEST2018.2572

Conference paper

Kounta, L., X. Capet, J. Jouanno, N. Kolodziejczyk, S. Ndoye, J. Deshayes. December 2016. West African upwelling waters and the circulation in the Northeast Tropical Atlantic.International Conference AWA "ICAWA 2015", https://www.documentation.ird.fr/hor/fdi:010067498 , Page 25, Dakar, Senegal

Remi M., **Kounta L**., Moussa D., Gaëlle D. C., Alban L. (2014), Atmospheric responses to anomalies of SST seasonal variations in the Northeastern Tropical Atlantic, EGU General Assembly 2014, Apr 2014, Vienna, Austria. pp.id.16854, 2014. https://ui.adsabs.harvard.edu/abs/2014EGUGA..1616854M

Data products

Kounta, L., P. Zarnetske, and L. Luo. 2025. Supporting data for "Climate Intervention through Stratospheric Aerosol Injection may partially mitigate marine heatwaves" ver 2. Environmental Data Initiative. https://doi.org/10.6073/pasta/61e0b03682967f2882c899d204702276

Youngflesh, C. K. Kapsar, A. Uscanga, P.J. Williams, J.W. Doser, **L. Kounta**, and P.L. Zarnetske. 2024. Inter- and intra-annual temperature and precipitation variability (1950-2022) across the ranges of non- migratory birds and their association with generation length version 4. Environmental Data Initiative. https://portal.edirepository.org/nis/mapbrowse?scope=edi&identifier=1623&revision=4

Policy bref

Hermes, J., C. Reason, **L. Kounta**, P. Nying'uro, P. Sagero, V. Ongoma, B. Atsu Foli, S. Osima, D. Mawren, R. Folorunsho, T. Duba, A. Mavume, A. Bié. 2021. Policy Brief 10: Extremes, Abrupt Changes and Managing Risks https://www.researchgate.net/publication/363915007_Policy_Brief_10_Extremes_Abrupt_Changes_and_Managing_Risks

Presentations

Kapsar, K., **Kounta, L**., Grady, J., Bills, P., Record, S., Strecker, A., Baiser, B., Zarnetske, P. (16 September 2025). Multi-scale environmental geodiversity and climate data layers for the National

Ecological Observatory Network (NEON) analyses across scales [Invited lecture]. University of Illinois at Chicago, Chicago, Illinois.

Kapsar, K., **Kounta, L**., Grady, J., Bills, P., Record, S., Strecker, A., Baiser, B., Zarnetske, P. (12 August 2025). Multi-scale environmental geodiversity and climate data layers for the National Ecological Observatory Network (NEON) analyses across scales [Conference presentation]. Ecological Society of America, Baltimore, Maryland

Lala Kounta, Lifeng Luo, Phoebe Zarnetske, October 2024. Marine heat waves under climate change and climate intervention, Fall Student Research Symposium Michigan State University, East Lansing, Michigan USA. Oral presentation

Lala Kounta, Gouri Anil, Daniel Martin Hueholt, Cheryl S Harrison, Amadou Thierno Gaye, Phoebe, Zarnetske, February 2024. Can solar radiation modification mitigate future increases in marine heatwaves? Ocean Sciences Meeting, New Orleans, Louisiana, USA. Poster

Shepard, I.D., A. Yue, **L. Kounta**, B. Baiser, A. Strecker, P.L. Zarnetske, S. Record. August 2023. Effects of linear disturbance on biodiversity across multiple taxonomic groups. Ecological Society of America Meeting. Portland, OR, USA. Oral presentation.

McGrew, A., C. Ganz, **L. Kounta**, A. Strecker, B. Baiser, S. Record, P.L. Zarnetske. August 2023. Size-based intraspecific trait variation-diversity relationships in freshwater fish across National Ecological Observatory Network (NEON) streams and lakes. Ecological Society of America Meeting. Portland, OR, USA. Oral presentation.

Kounta,L. July 2023. Climate Change, Climate Intervention & Marine Heat Waves. African Futures Dissemination Workshop, Michigan State University, East Lansing, MI, USA. Oral presentation

Kounta, L. May 25 2023. Climate Variability and Marine Heat Waves Under Climate Change and Climate Intervention Scenarios: From Regional to Global Scales. Invited. National Center for Atmospheric Research (NCAR), Boulder, CO, USA.

Kounta, L. March 2020. The circulation in the Northeastern Tropical Atlantic: seasonal variability and dynamics of the West African poleward boundary current (WABC), The Canary Current Eastern Boundary Upwelling System. Mindelo, Cabo Verde.

Kounta, L. February 2020 Heat Waves Along the West African Seaboard: Role of the West African Boundary Current in Present and Future Conditions. Climate Science workshop . AIMS-Rwamda, Kigali, Rwanda.

Ndoye S, **L. Kounta**, A. Sylla, X. Capet, J. Mignot V. Echevin, M. Diakhaté, E. Machu. October 2019. SCUS-2050: Exploring the future of the Southern Canary Upwelling. African Climate Risks Conference. Addis Abiba, Ethiopia.

Kounta, L., Capet, X., Jouanno, J., Kolodziejczyk, N., Sow, B., Gaye, A. T. September 2019. Étude du cycle saisonnier et de la dynamique du courant de bord Ouest Africain. Tropical Atlantic Climate and Coastal Variability (TACCOVAR) conference**,** Cotonou, Bénin.

Kounta, L., Capet, X., Jouanno, J., Kolodziejczyk, N., Sow, B., Gaye, A. T. May 3-5, 2018. Etude de la variabilité saisonnière et de la dynamique du West African poleward Boundary Current (WABC). Casaman Sun VIth Edition. Ziguinchor, Senegal.

Ndoye S, X. Capet, V. Echevin, E. Machu, **L. Kounta**, A. Sylla, J. Mignot, M. Diakhaté and A.T. Gaye. September 2017. SCUS-2050: Exploring the future of the Southern Canary Upwelling System. WP4 Downscaling in forced high resolution SCUS simulations. Future Climate For Africa MidTerm Conference. Cape Town, South Africa.

PROFESSIONAL AFFILIATIONS

African Meteorological Society Awards committee, Society for Women in Marine Science

Association of Climate Change Officers, American Geosciences Union